

ELITE IGCSE ACADEMY

Private Past Paper Solutions

Jan 2021 P1HR Worked Solutions

Jan 2021 | P1HR

25 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3**TOPIC: **Fractions**

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1 Show that $3\frac{1}{5} \times 1\frac{5}{6} = 5\frac{13}{15}$

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION**Q#: 1****Marks: 3**TOPIC: **Fractions**

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WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Fractions. The tag is correct.**Method**

Convert the mixed numbers to improper fractions:

$$3\frac{1}{5} = \frac{16}{5}$$

$$1\frac{5}{6} = \frac{11}{6}$$

Now multiply:

$$\begin{aligned}\frac{16}{5} \times \frac{11}{6} &= \frac{176}{30} \\ &= \frac{88}{15} = 5\frac{13}{15}\end{aligned}$$

FINAL ANSWER

$$5\frac{13}{15}$$

PAST PAPER QUESTION**Q#: 2****Marks: 4**TOPIC: **Statistics Toolkit**

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2 Given that $a < b < c$

the four whole numbers a, a, b and c have

a mode of 7

a median of 8.5

a mean of 9

Work out the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION**Q#: 2****Marks: 4**TOPIC: **Statistics Toolkit**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Statistics Toolkit. The tag is correct.**Solution**The mode is 7, and the numbers are a, a, b, c , so

$$a = 7$$

The median of four numbers is

$$\frac{a + b}{2}$$

So

$$\frac{7 + b}{2} = 8.5$$

$$7 + b = 17$$

$$b = 10$$

The mean is 9, so the total is

$$4 \times 9 = 36$$

$$7 + 7 + 10 + c = 36$$

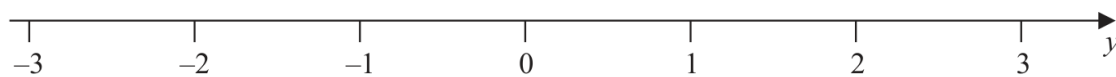
$$c = 12$$

FINAL ANSWER $a = 7, b = 10, c = 12.$

PAST PAPER QUESTION**Q#: 3****Marks: 4****TOPIC: Solving Inequalities**

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- 3 (a) On the number line, show the inequality $-2 \leq y < 1$



(2)

n is an integer.

- (b) Write down all the values of n that satisfy $-3.4 < n \leq 2$

.....
(2)

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 4****TOPIC: Solving Inequalities**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Solving inequalities. The tag is correct.**Solution**

(a) For

$$-2 \leq y < 1$$

use a closed circle at -2 , an open circle at 1 , and shade the line between them.

(b)

$$-3.4 < n \leq 2$$

The integer values are

$$-3, -2, -1, 0, 1, 2$$

FINAL ANSWERClosed at -2 , open at 1 ; integers $-3, -2, -1, 0, 1, 2$.

PAST PAPER QUESTION**Q#: 4****Marks: 3****TOPIC: Standard & Compound Units**

Jan 2021 P1HR - Jan 2021 | P1HR

- 4 A train journey from Paris to Amsterdam took 3 hours 24 minutes.
The total distance the train travelled was 433.5 km.

Work out the average speed of the train.
Give your answer in kilometres per hour.

..... km/h

(Total for Question 4 is 3 marks)

PAST PAPER SOLUTION**Q#: 4** **Marks: 3**TOPIC: **Standard & Compound Units**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to Standard & Compound Units.**Solution**

The time is

$$3 \text{ h } 24 \text{ min} = 3.4 \text{ h}$$

Average speed:

$$433.5 \div 3.4 = 127.5$$

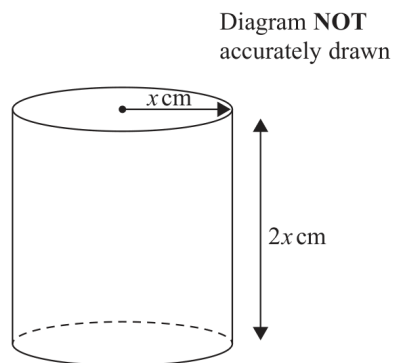
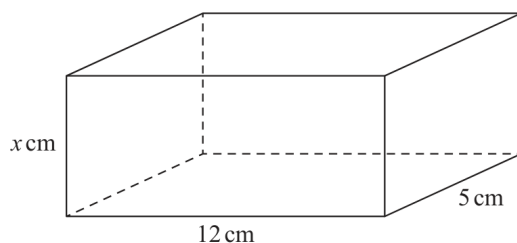
FINAL ANSWER

127.5 km/h.

PAST PAPER QUESTION**Q#: 5** **Marks: 4****TOPIC: Volume & Surface Area**

Jan 2021 P1HR - Jan 2021 | P1HR

- 5 The diagram shows a cuboid and a cylinder.



The dimensions of the cuboid are x cm by 12 cm by 5 cm.
The volume of the cuboid is 270 cm^3

The radius of the cylinder is x cm.
The height of the cylinder is $2x$ cm.

- (a) Work out the volume of the cylinder.
Give your answer correct to the nearest whole number.

..... cm^3
(3)

- (b) Change 1 m^3 to cm^3

..... cm^3
(1)

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 4****TOPIC: Volume & Surface Area**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Volume & Surface Area. The tag is correct.**Solution**

For the cuboid,

$$12 \times 5 \times x = 270$$

$$60x = 270$$

$$x = 4.5$$

The cylinder has radius 4.5 cm and height 9 cm.

$$V = \pi r^2 h = \pi (4.5)^2 (9) = 572.555 \dots$$

Also,

$$1 \text{ m}^3 = 100^3 = 1000000 \text{ cm}^3$$

FINAL ANSWER573 cm³, and 1 m³ = 1000000 cm³.

PAST PAPER QUESTION**Q#: 6****Marks: 5****TOPIC: Rearranging Formulas**

Jan 2021 P1HR - Jan 2021 | P1HR

- 6 (a) Make c the subject of $A = \frac{c}{y} - 5z$

.....
(2)

- (b) Write down the value of g^0

.....
(1)

- (c) Factorise $x^2 - 11x + 24$

.....
(2)

.....
(Total for Question 6 is 5 marks)

PAST PAPER SOLUTION**Q#: 6** **Marks: 5**TOPIC: **Rearranging Formulas**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Rearranging formulas. The tag is acceptable.**Solution**

(a)

$$A = \frac{c}{y} - 5z$$

$$A + 5z = \frac{c}{y}$$

$$c = y(A + 5z)$$

(b) $g^0 = 1$.

(c)

$$x^2 - 11x + 24 = (x - 3)(x - 8)$$

FINAL ANSWER

$$c = y(A + 5z), 1, (x - 3)(x - 8)$$

PAST PAPER QUESTION**Q#: 7****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jan 2021 P1HR - Jan 2021 | P1HR

- 7 Kuro invests 50 000 yen for 3 years in a savings account.
She gets 2.4% per year compound interest.

Work out how much money Kuro will have in her savings account at the end of the 3 years.
Give your answer correct to the nearest yen.

..... yen

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION**Q#: 7****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Compound Interest and Depreciation. The tag is correct.**Method**

The multiplier for a 2.4% increase is

$$1.024$$

After 3 years,

$$50000(1.024)^3 = 53687.0912$$

Correct to the nearest yen,

$$53687.0912 \approx 53687$$

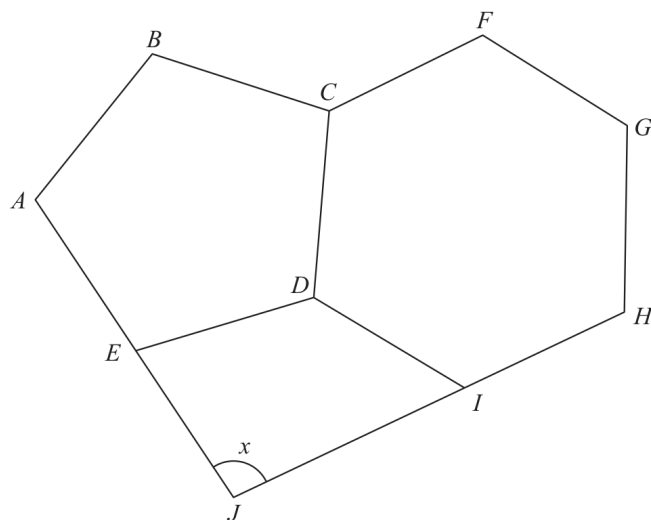
FINAL ANSWER

53687 yen

PAST PAPER QUESTION**Q#: 8****Marks: 5****TOPIC: Angles in Polygons & Parallel Lines**

Jan 2021 P1HR - Jan 2021 | P1HR

- 8 The diagram shows a regular pentagon, $ABCDE$, a regular hexagon, $CFGHID$, and a quadrilateral, $EDIJ$.

Diagram **NOT** accurately drawn

AEJ and HIJ are straight lines.

Work out the size of the angle marked x .
Show your working clearly.

(Total for Question 8 is 5 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 5****TOPIC: Angles in Polygons & Parallel Lines**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to angles in polygons and parallel lines. The question uses regular polygon angles.

Solution

A regular pentagon has interior angle

$$\frac{(5 - 2)180}{5} = 108^\circ$$

A regular hexagon has interior angle

$$\frac{(6 - 2)180}{6} = 120^\circ$$

At E , since AEJ is a straight line,

$$\angle DEJ = 180 - 108 = 72^\circ$$

At D , the angles around the point include the pentagon angle 108° and the hexagon angle 120° , so

$$\angle EDI = 360 - 108 - 120 = 132^\circ$$

At I , since HII is a straight line,

$$\angle DIJ = 180 - 120 = 60^\circ$$

In quadrilateral $EDIJ$, angles sum to 360° :

$$x = 360 - 72 - 132 - 60 = 96$$

FINAL ANSWER

96° .

PAST PAPER QUESTION**Q#: 9****Marks: 4****TOPIC: Prime Factors, HCF & LCM**

Jan 2021 P1HR - Jan 2021 | P1HR

9 $A = 2^8 \times 3^5 \times 11^4$ $B = 2^6 \times 3 \times 11^8$

(a) Find the highest common factor (HCF) of A and B .

.....
(2)

(b) Find the lowest common multiple (LCM) of $2A$ and $3B$.
Give the LCM as a product of powers of its prime factors.

.....
(2)

.....
(Total for Question 9 is 4 marks)

PAST PAPER SOLUTION**Q#: 9****Marks: 4****TOPIC: Prime Factors, HCF & LCM**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Prime factors, HCF and LCM. The tag is correct.**Method**

$$A = 2^8 \times 3^5 \times 11^4$$

$$B = 2^6 \times 3 \times 11^8$$

For the HCF, take the smaller power of each common prime:

$$\text{HCF} = 2^6 \times 3 \times 11^4$$

Now

$$2A = 2^9 \times 3^5 \times 11^4$$

and

$$3B = 2^6 \times 3^2 \times 11^8$$

For the LCM of $2A$ and $3B$, take the larger power of each prime:

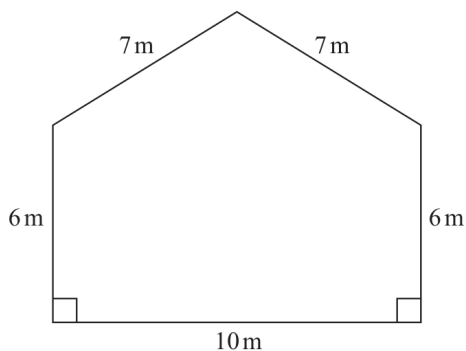
$$\text{LCM} = 2^9 \times 3^5 \times 11^8$$

FINAL ANSWER(a) $2^6 \times 3 \times 11^4$, (b) $2^9 \times 3^5 \times 11^8$

PAST PAPER QUESTION**Q#: 10****Marks: 5****TOPIC: Area & Perimeter**

Jan 2021 P1HR - Jan 2021 | P1HR

- 10** The diagram shows one face of a wall.
This face is in the shape of a pentagon with exactly one line of symmetry.



Omondi is going to paint this face of the wall once.
He has to buy all the paint that he needs to use.

The paint in each tin of paint Omondi is going to buy will cover 16m^2 of the face of the wall.

Work out the least number of tins of paint Omondi will need to buy.
Show your working clearly.

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 5**TOPIC: **Area & Perimeter**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to Area & Perimeter.**Solution**

The wall is a rectangle plus an isosceles triangle.

Rectangle area:

$$10 \times 6 = 60$$

For the triangle, half the base is 5 m.

$$h^2 = 7^2 - 5^2 = 24$$

$$h = \sqrt{24}$$

Triangle area:

$$\frac{1}{2} \times 10 \times \sqrt{24} = 24.494 \dots$$

Total area:

$$60 + 24.494 \dots = 84.494 \dots$$

Each tin covers 16 m^2 :

$$\frac{84.494 \dots}{16} = 5.28 \dots$$

Omondi must buy whole tins, so he needs 6 tins.

FINAL ANSWER

6 tins.

PAST PAPER QUESTION**Q#: 11****Marks: 6****TOPIC: Cumulative Frequency Diagrams**

Jan 2021 P1HR - Jan 2021 | P1HR

- 11 The manager of a call centre asked the 120 people, who rang the call centre last week, how long they each waited before their call was answered.

The table gives information about their replies.

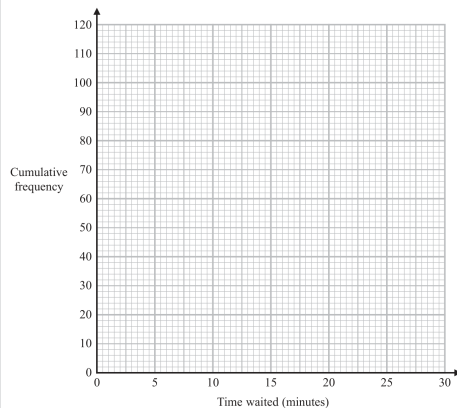
Time waited (t minutes)	Frequency
$0 < t \leq 5$	8
$5 < t \leq 10$	15
$10 < t \leq 15$	17
$15 < t \leq 20$	28
$20 < t \leq 25$	33
$25 < t \leq 30$	19

- (a) Complete the cumulative frequency table.

Time waited (t minutes)	Cumulative frequency
$0 < t \leq 5$	
$0 < t \leq 10$	
$0 < t \leq 15$	
$0 < t \leq 20$	
$0 < t \leq 25$	
$0 < t \leq 30$	

(1)

- (b) On the grid below, draw a cumulative frequency graph for your table.



(2)

- (c) Use your graph to find an estimate for the median of the times waited.

_____ minutes
(1)

- (d) Using your graph, find an estimate for the percentage of the 120 people who said that they waited longer than 23 minutes before their call was answered. Show your working clearly.

_____ %
(2)

(Total for Question 11 is 6 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 6****TOPIC: Cumulative Frequency Diagrams**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Cumulative Frequency Diagrams. The tag is correct.**Solution**

Complete the cumulative frequencies:

8, 23, 40, 68, 101, 120

Plot the points

 $(0, 0), (5, 8), (10, 23), (15, 40), (20, 68), (25, 101), (30, 120)$

The median is at cumulative frequency 60.

This lies between (15, 40) and (20, 68):

$$15 + \frac{60 - 40}{68 - 40} \times 5 = 18.57 \dots$$

So the median is about 18.6 minutes.

At 23 minutes:

$$68 + \frac{3}{5}(101 - 68) = 87.8$$

Number who waited longer than 23 minutes:

$$120 - 87.8 = 32.2$$

$$\frac{32.2}{120} \times 100 = 26.8\%$$

FINAL ANSWER

cumulative frequencies 8, 23, 40, 68, 101, 120, median about 18.6 minutes, percentage about 27%.

PAST PAPER QUESTION**Q#: 12****Marks: 9**TOPIC: **Algebraic Roots & Indices**

Jan 2021 P1HR - Jan 2021 | P1HR

12 (a) Simplify $(16e^{10}f^6)^{\frac{1}{2}}$
(2)(b) Write $\frac{2x+1}{4} + \frac{x-2}{3}$ as a single fraction in its simplest form......
(3)Given that $4^{k+3} = 16 \times 2^k$ (c) find the value of k .
Show your working clearly. $k =$
(4)**(Total for Question 12 is 9 marks)**

PAST PAPER SOLUTION**Q#: 12****Marks: 9****TOPIC: Algebraic Roots & Indices**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to Algebraic Roots & Indices.**Solution**

(a)

$$(16e^{10}f^6)^{\frac{1}{2}} = 4e^5f^3$$

(b)

$$\begin{aligned}\frac{2x+1}{4} + \frac{x-2}{3} &= \frac{3(2x+1) + 4(x-2)}{12} \\ &= \frac{6x+3+4x-8}{12} = \frac{10x-5}{12} = \frac{5(2x-1)}{12}\end{aligned}$$

(c)

$$4^{k+3} = 16 \times 2^k$$

$$(2^2)^{k+3} = 2^4 \times 2^k$$

$$2^{2k+6} = 2^{k+4}$$

$$2k+6 = k+4$$

$$k = -2$$

FINAL ANSWER(a) $4e^5f^3$, (b) $\frac{5(2x-1)}{12}$, (c) $k = -2$.

PAST PAPER QUESTION**Q#: 13****Marks: 2**TOPIC: **Vectors**

Jan 2021 P1HR - Jan 2021 | P1HR

13 Here are two vectors.

$$\overrightarrow{AB} = \begin{pmatrix} 5 \\ 3 \end{pmatrix} \quad \overrightarrow{CB} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$$

Find, as a column vector, $\overrightarrow{AC}$

(Total for Question 13 is 2 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 2**TOPIC: **Vectors**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Vectors. The tag is correct.**Solution**

$$\overrightarrow{AB} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}$$

and

$$\overrightarrow{CB} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$$

So

$$\overrightarrow{BC} = -\overrightarrow{CB} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$$

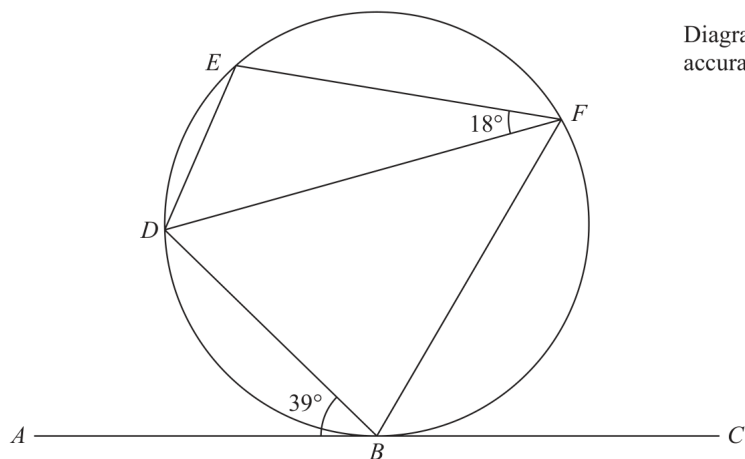
$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = \begin{pmatrix} 5 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 \\ -4 \end{pmatrix} = \begin{pmatrix} 7 \\ -1 \end{pmatrix}$$

FINAL ANSWER

$$\begin{pmatrix} 7 \\ -1 \end{pmatrix}.$$

PAST PAPER QUESTION**Q#: 14** **Marks: 4**TOPIC: **Circle Theorems**

Jan 2021 P1HR - Jan 2021 | P1HR

14Diagram **NOT**
accurately drawn B, D, E and F are points on a circle. ABC is the tangent at B to the circle.Angle $ABD = 39^\circ$ Angle $EFD = 18^\circ$ Work out the size of angle BDE .

Give reasons for your working.

(Total for Question 14 is 4 marks)

PAST PAPER SOLUTION**Q#: 14****Marks: 4****TOPIC: Circle Theorems**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Circle Theorems. The tag is correct.**Solution**

By the alternate segment theorem, the angle between the tangent and chord BD is equal to the angle in the opposite segment:

$$\angle BFD = 39^\circ$$

Given

$$\angle EFD = 18^\circ$$

So

$$\angle BFE = 39^\circ + 18^\circ = 57^\circ$$

Opposite angles in cyclic quadrilateral $BDEF$ add to 180° :

$$\angle BDE = 180^\circ - 57^\circ = 123^\circ$$

FINAL ANSWER

123°.

PAST PAPER QUESTION**Q#: 15****Marks: 5**TOPIC: **Algebraic Proof**

Jan 2021 P1HR - Jan 2021 | P1HR

15 (a) Use algebra to show that $4.\dot{5}\dot{7} = 4\frac{19}{33}$

(b) Show that $\frac{2}{6-3\sqrt{2}}$ can be written in the form $\frac{a+\sqrt{a}}{b}$
 where a and b are integers.
 Show your working clearly.

(2)

(3)

(Total for Question 15 is 5 marks)

PAST PAPER SOLUTION**Q#: 15****Marks: 5****TOPIC: Algebraic Proof**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Algebraic proof. The tag is correct.**Solution**

(a) Let

$$x = 4.\dot{5}\dot{7}$$

Then

$$100x = 457.\dot{5}\dot{7}$$

Subtract:

$$100x - x = 457.\dot{5}\dot{7} - 4.\dot{5}\dot{7}$$

$$99x = 453$$

$$x = \frac{453}{99} = \frac{151}{33} = 4\frac{19}{33}$$

(b)

$$\begin{aligned} \frac{2}{6-3\sqrt{2}} \times \frac{6+3\sqrt{2}}{6+3\sqrt{2}} &= \frac{12+6\sqrt{2}}{36-18} \\ &= \frac{12+6\sqrt{2}}{18} = \frac{2+\sqrt{2}}{3} \end{aligned}$$

So this is in the form $\frac{a+\sqrt{a}}{b}$, with $a=2$ and $b=3$.**FINAL ANSWER**

$$4.\dot{5}\dot{7} = 4\frac{19}{33} \text{ and } \frac{2}{6-3\sqrt{2}} = \frac{2+\sqrt{2}}{3}.$$

PAST PAPER QUESTION**Q#: 16****Marks: 5****TOPIC: Expanding Brackets**

Jan 2021 P1HR - Jan 2021 | P1HR

16 (a) Expand and simplify $(x + 4)(x - 2)(x + 1)$

.....
(3)

(b) Express $x^2 - 10x + 40$ in the form $(x + a)^2 + b$, where a and b are integers.

.....
(2)

(Total for Question 16 is 5 marks)

PAST PAPER SOLUTION**Q#: 16****Marks: 5****TOPIC: Expanding Brackets**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to expanding brackets. The expansion is the largest part of the question.

Solution

(a)

$$(x + 4)(x - 2)(x + 1)$$

$$(x - 2)(x + 1) = x^2 - x - 2$$

$$(x + 4)(x^2 - x - 2) = x^3 + 3x^2 - 6x - 8$$

(b)

$$x^2 - 10x + 40 = (x - 5)^2 + 15$$

FINAL ANSWER(a) $x^3 + 3x^2 - 6x - 8$, (b) $(x - 5)^2 + 15$

PAST PAPER QUESTION**Q#: 17****Marks: 5****TOPIC: Simultaneous Equations**

Jan 2021 P1HR - Jan 2021 | P1HR

17 Solve the simultaneous equations

$$x - 6y = 5$$

$$xy - 2y^2 = 6$$

Show clear algebraic working.

(Total for Question 17 is 5 marks)

PAST PAPER SOLUTION**Q#: 17****Marks: 5****TOPIC: Simultaneous Equations**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Simultaneous equations. The tag is correct.**Solution**

$$x - 6y = 5$$

$$xy - 2y^2 = 6$$

From the first equation:

$$x = 6y + 5$$

Substitute into the second equation:

$$y(6y + 5) - 2y^2 = 6$$

$$4y^2 + 5y - 6 = 0$$

Factorise:

$$(4y - 3)(y + 2) = 0$$

So

$$y = \frac{3}{4} \quad \text{or} \quad y = -2$$

If $y = \frac{3}{4}$:

$$x = 6\left(\frac{3}{4}\right) + 5 = \frac{19}{2}$$

If $y = -2$:

$$x = 6(-2) + 5 = -7$$

FINAL ANSWER

$$(x, y) = \left(\frac{19}{2}, \frac{3}{4}\right) \text{ or } (-7, -2).$$

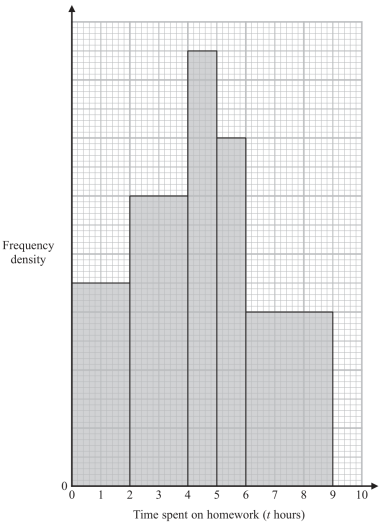
PAST PAPER QUESTION

Q#: 18 Marks: 5

TOPIC: Histograms

Jan 2021 P1HR - Jan 2021 | P1HR

18 The histogram and the table give some information about the amounts of time, in hours, that Year 11 students at Bergdesh Academy spent, in total, on their homework last week. No student in Year 11 spent longer than 9 hours on their homework.



Time spent on homework (<i>t</i> hours)	Frequency
$0 < t \leq 2$	28
$2 < t \leq 4$	
$4 < t \leq 5$	
$5 < t \leq 6$	
$6 < t \leq 9$	

Using the information in the histogram and in the table, work out an estimate for the mean amount of time the Year 11 students spent on their homework last week. Give your answer in hours correct to 3 significant figures.

_____ hours

(Total for Question 18 is 5 marks)

PAST PAPER SOLUTION**Q#: 18****Marks: 5****TOPIC: Histograms**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Histograms. The tag is correct.**Solution**

The $0 < t \leq 2$ class has frequency 28. From the histogram, the relative heights are

$$1, 4, 9, 6, 1$$

So the frequencies are

$$28, 112, 126, 84, 42$$

Use midpoints 1, 3, 4.5, 5.5, 7.5.

$$\text{total frequency} = 28 + 112 + 126 + 84 + 42 = 392$$

$$\text{estimated total} = 1(28) + 3(112) + 4.5(126) + 5.5(84) + 7.5(42) = 1708$$

$$\text{mean} = \frac{1708}{392} = 4.357 \dots$$

FINAL ANSWER

4.36 hours.

PAST PAPER QUESTION**Q#: 19****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Jan 2021 P1HR - Jan 2021 | P1HR

19 $k = \frac{t}{a - h}$

 $t = 14$ correct to 2 significant figures $a = 7.8$ correct to 2 significant figures $h = 3.4$ correct to 2 significant figures

Work out the lower bound for the value of k .
Show your working clearly.

.....
(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION**Q#: 19****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Rounding, Estimation and Bounds. The tag is correct.**Method**

$$k = \frac{t}{a - h}$$

For the lower bound of k , use the lower bound of t and make the denominator as large as possible.

$$t \geq 13.5$$

$$a < 7.85, \quad h \geq 3.35$$

$$(a - h)_{\text{upper}} = 7.85 - 3.35 = 4.5$$

$$k_{\text{lower}} = \frac{13.5}{4.5} = 3$$

FINAL ANSWER**3**

PAST PAPER QUESTION**Q#: 20****Marks: 5****TOPIC: Differentiation**

Jan 2021 P1HR - Jan 2021 | P1HR

20 A particle P is moving along a straight line.The fixed point O lies on the line.At time t seconds ($t \geq 0$), the displacement of P from O is s metres where

$$s = t^3 - 9t^2 + 33t - 6$$

Find the minimum speed of P .

..... m/s

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 5****TOPIC: Differentiation**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved from Linear Graphs to Differentiation.**Method**

$$s = t^3 - 9t^2 + 33t - 6$$

Velocity is the rate of change of displacement:

$$v = \frac{ds}{dt}$$

$$v = 3t^2 - 18t + 33$$

Complete the square:

$$v = 3(t^2 - 6t + 11)$$

$$v = 3[(t - 3)^2 + 2]$$

$$v = 3(t - 3)^2 + 6$$

Since $(t - 3)^2 \geq 0$, the minimum value of v is

$$6$$

The velocity is always positive, so the minimum speed is also 6.

FINAL ANSWER

6 m/s

PAST PAPER QUESTION**Q#: 20****Marks: 5****TOPIC: Differentiation**

Jan 2021 P1HR - Jan 2021 | P1HR

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PAST PAPER QUESTION**Q#: 21****Marks: 6****TOPIC: Sequences**

Jan 2021 P1HR - Jan 2021 | P1HR

- 21 The n th term of an arithmetic series is u_n where $u_n > 0$ for all n
The sum to n terms of the series is S_n

Given that $u_4 = 6$ and that $S_{11} = (u_4)^2 + 18$

find the value of u_{20}

(Total for Question 21 is 6 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 6****TOPIC: Sequences**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**Let $u_6 = v$.

For an arithmetic series,

$$S_{11} = \frac{11}{2}(u_1 + u_{11})$$

The average of u_1 and u_{11} is u_6 , so

$$S_{11} = 11u_6 = 11v$$

Given

$$S_{11} = (u_6)^2 + 18$$

$$11v = v^2 + 18$$

$$v^2 - 11v + 18 = 0$$

$$(v - 2)(v - 9) = 0$$

So

$$u_6 = 2 \quad \text{or} \quad u_6 = 9$$

But $u_4 = 6$. If $u_6 = 2$, then the common difference is negative and later terms would not all be positive. Therefore,

$$u_6 = 9$$

$$d = \frac{u_6 - u_4}{2} = \frac{9 - 6}{2} = \frac{3}{2}$$

$$u_{20} = u_6 + 14d$$

$$u_{20} = 9 + 14\left(\frac{3}{2}\right) = 30$$

FINAL ANSWER

$$u_{20} = 30$$

PAST PAPER QUESTION**Q#: 21****Marks: 6****TOPIC: Sequences**

Jan 2021 P1HR - Jan 2021 | P1HR

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(Total for Question 21 is 6 marks)

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Jan 2021 P1HR - Jan 2021 | P1HR

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FINAL ANSWER

$$u_{20} = 30$$

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Coordinate Geometry**

Jan 2021 P1HR - Jan 2021 | P1HR

22 ABC is an isosceles triangle with $AB = AC$.

B is the point with coordinates $(-1, 5)$

C is the point with coordinates $(2, 10)$

M is the midpoint of BC .

Find an equation of the line through the points A and M .

Give your answer in the form $py + qx = r$ where p , q and r are integers.

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 5****TOPIC: Coordinate Geometry**

Jan 2021 P1HR - Jan 2021 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**Since $AB = AC$, point A lies on the perpendicular bisector of BC .The midpoint of BC is

$$M = \left(\frac{-1 + 2}{2}, \frac{5 + 10}{2} \right)$$

$$M = \left(\frac{1}{2}, \frac{15}{2} \right)$$

The gradient of BC is

$$\frac{10 - 5}{2 - (-1)} = \frac{5}{3}$$

So the gradient of the perpendicular bisector is

$$-\frac{3}{5}$$

Use the line through M :

$$y - \frac{15}{2} = -\frac{3}{5} \left(x - \frac{1}{2} \right)$$

Multiply by 10:

$$10y - 75 = -6x + 3$$

$$10y + 6x = 78$$

Divide by 2:

$$5y + 3x = 39$$

FINAL ANSWER

$$5y + 3x = 39$$

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